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IF3E Carpooling website project report



Introduction :

The IF3E UE was designed to teach students the basics of the relational databses and this mid-semester project was here to help students gather all the stuff they learnt during this half-semester in one big project.

Working on this project, I had to design the architecture of a database, I had to write the SQL resquests needed to gather the information and I had to write the PHP needed in order to put together the power of the relational databases with a web browser.

All these skills were completly new for me so I had to work two weeks on this website in order to use all the skills I’ve learnt during the lectures.

In this report, you will see the different sides of the project such as the relational model used or the implemented features.

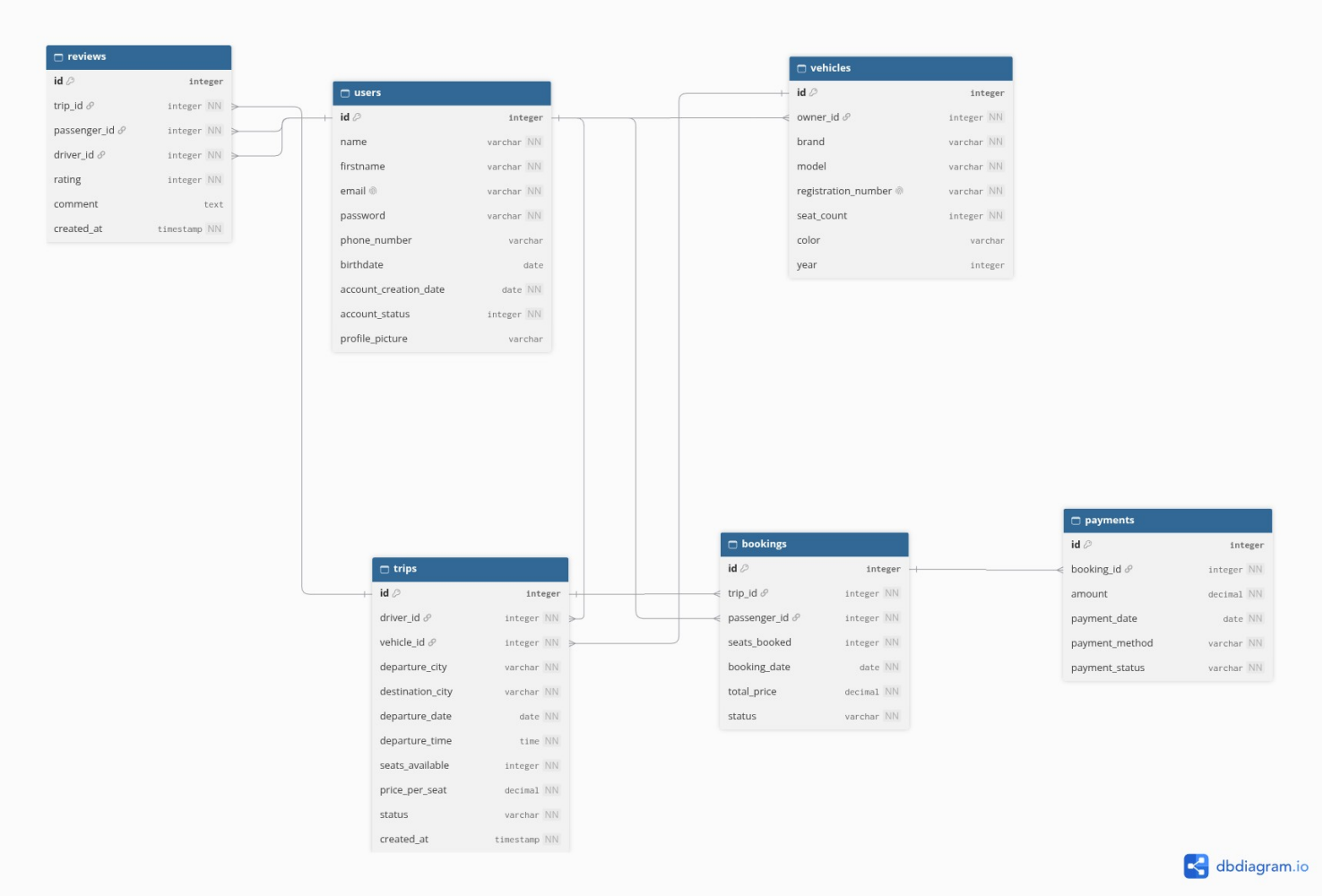
Table of contents :

Table of contents

Introduction :..... 2
Table of contents :.....3
The entity-relationship diagram and the relational model.....4
Explanation of 3NF normalization.....6
Implemented features.....7
Conclusion.....8

The entity-relationship diagram and the relational model

Here is the entity-relationship diagram used to represent the different relations between the tables in the project. This diagram has been created with dbdiagram.io :



Since the picture is quite small, you'll have to zoom in order to see the relational model correctly.

Basically, the table named users will contain all the infromation required about the users and all the information needed to login.

Some users may be drivers, that's the reason why the table users is connected with the table vehicles which contains the information required about the vehicles posessed by the users which will be used for carpooling.

Next, there is the trips table. When the users want to drive somewhere, the may decide to invite other users of the website to join them. They will need to create a new trip specifiying which vehicle will be used for the trip, and other information about this one (What is the departure city and the destination city ? When will it be ? …).

Next, an interested passenger will probably book the trip. That’s the reason why the bookings table is here. This table will contain the information about the trip selected and the passenger who wanted to book the trip.

The passenger will need to pay for its booking and this payment will be recorded in the payments table with some information about the payment such as the method used.

Finally, once the trip is over, a passenger may eventually leave a comment on the website about the driver, which will be saved in the reviews table alongside with the rating and the ids of the passenger and the driver.

Explanation of 3NF normalization

Implemented features

This project includes the basic required features asked in order to respect the instructions.

We can basically use the website for creating new accounts by using the registering option and log in to them.

We can also use our accounts to add the cars we posess and create trips in order to publish them to the other users of the website.

Moreover, there are also a few optional features that were not required in the project but are still present in the website such as an international telephone format helper tool in the registering page

Conclusion

The basic idea of the UE IF3E was to teach how to use relational databases and how to interact with them. Even though there are a lot of possible use in a programmer life, the exercises still stayed a bit abstract and were not totally enjoyable since the purpose of the TD's exercise was only to help us practicing what we learnt in course.

On the other hand, this project tought us relational databases, SQL and PHP in a completly other way ; I had to work in a specific environment with a specific context and creating a full website.

To be honest, I found this project very difficult, not only because I never saw how to use PHP and how to connect it with SQL before, but because of the small amount of time given too. I only had two weeks to do the project, which was quite stressful I must admit but also very rewarding too.

There is a concept called the Parkinson's law which state that the work we do fill always the amount of time we're given to do it… because I had to do this project in a very small amount of time, I had the feeling that I put great efforts on this project, even if I had to ask help on the internet as every good programmer (stack overflow, AI, w3schools…) all that stuff helped me and I can proudly announce that I created a website in two weeks.

I want to conclude this report saying that I'm quite proud of me even if I am sure there is a mountain of optimization and other cool new features I could add to the website, I could for instance add a ban-system for admins or a chat system for the users, but time lacked so I decided to go only right to the goal and deliver the best website I could possibly submit in two weeks.

I hope that you will enjoy my brand new website !